

Inverter Cooling Solution



Background:

As summer approaches, the average daily light hours increase. When the efficiency of photovoltaic power generation grows, the rising temperatures also have an impact on the inside of the inverters. The interior of the inverter consists of highly sophisticated components, which have a high demanding in terms of temperature. In summer, as the intensity of sunlight increases, the heat transferred to the inverter shell through solar radiation also increases, causing the casing temperature to rise.

Cause Analysis:

Why Do Inverter Generate Heat?

After the inverter starts working, all parts of its internal components begin to run and the power increases, generating a large amount of heat. **This heat is transferred to the inverter shell by means of heat conduction and heat convection, causing the temperature of the shell to rise.**

Why Is Heat Dissipation Necessary for Inverters?

The proper functioning of the inverter relies on the internal components operating in the allowed temperature range. The influence of temperature on inverter components is reflected in two aspects. On the one hand, temperature affects the material properties. Electronic components show different parameter characteristics in different temperature environment. Once the change of component parameters exceeds the predetermined range, circuit failure may occur. On the other hand, the structural stability of electronic components is also affected by temperature. Under high or low temperature environment, the internal structure of electronic components may change, for example, some materials will undergo physical deformation under high temperature, which may cause damage to the components, making the inverter unable to work normally.

Inverter Heat Dissipation Design:

Nowadays, common inverter cooling methods mainly include liquid cooling, air cooling and natural cooling. For low power inverters such as X1-Boost-G4, aluminum heat sink is a good choice. The heat sink increases the surface area of heat exchange, allowing the air exchanging heat with the surface of the heat sink. When the heat was taken away, the inverter can have a relatively proper interior environment. For high power models such as X3-Hybrid-G4, Solax has equipped a cooling fan. When the fan is turned on, forced convection occurs in the inverter, which makes the heat flow and cools down the high temperature.



Figure 1: Aluminum heat sink



Figure 2: Cooling fan

The role of inverter housing in heat dissipation:

Aluminum heat-sink and fan can transfer the internal heat well, besides, the inverter case also plays an auxiliary role in the heat dissipation of the whole machine. The inverter housing is in close contact with the heat sink, and the heat concentrated in the heat sink can be transferred to the inverter housing by means of thermal conductivity. At the same time, the inverter shell also absorbs part of the heat transported in the form of thermal convection, which comes from the higher temperature air inside the inverter. After these two forms of heat are transferred to the shell, the shell exchanges heat with the lower temperature air outside the shell by convection to achieve the cooling effect.

Conclusion:

The components inside the inverter have high temperature requirements, which makes it necessary to maintain a certain temperature range inside the inverter. During the summer months, when outdoor temperatures increase, the overall temperature of the inverter increases accordingly, adding to the challenge of keeping the inverter functioning properly. SolaX inverters equipped with aluminum heat sinks and fans efficiently transfer heat through the shell to the external environment, ensuring that the inverter components will suffer less damages. Both of these above cooling methods are achieved with the inverter shell as the medium, therefore it is normal for the temperature to rise.